

ATHARVA ATUL REGE

Undergraduate Student | Machine Learning, Computer Vision, Generative and Agentic AI

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🏢 EXPERIENCE

AI and DevOps Engineering Intern, DevSecOps

May 2026 – July 2026

Bengaluru, India

Hewlett-Packard (HP Inc.)

- Shipped an agentic Jira triage workflow on Amazon Bedrock, orchestrating **4 specialized agents** to automate epic classification, reduce manual triage effort by **30%**, and run through an Azure DevOps pipeline in production.
- Integrated LLM agents with **3 live engineering systems** (GitHub, Jira, and Microsoft Teams) using AWS Strands SDK and custom GitHub Copilot agents, automating **5+** ticket-management operations.

Research Intern, advised by Prof. Imran Razzak

December 2025 – Present

Abu Dhabi, UAE

Mohamed bin Zayed University of Artificial Intelligence

- Co-authored **7 papers** (first author on 5), 4 accepted at MICCAI 2026 workshops: SCALA (oral, **top 15%**), BEAT, RADAR, and FieldFiLM, spanning cardiac segmentation, TB screening, and MRI field-strength adaptation.
- PRIMED (WACV 2027, under review): Reduced generative MRI synthesis from **4 networks and 1,000 sampling steps** to **1 network and 1 forward pass**, achieving **28.61 dB PSNR** while synthesizing each volume in **0.152 s**.
- DB-INR (WACV 2027, under review): Built a dual-band sinusoidal implicit neural representation for segmentation: **0.851 Dice** on OASIS-2D, **+0.039 Dice** across 11 MedSegBench datasets, **1.8× faster** and **1.5× leaner** inference.
- TuLaBM: Designed a latent bridge matching translator with tumour-biased attention for cross-domain medical image translation, improving tumour-region performance while maintaining **0.097 s/image** inference.
- Placed **4th of about 200 submissions** in the RARE 2026 challenge with AFA-LoRA, a **25-model** LoRA ensemble.

Student Researcher, advised by Prof. Jeny Rajan

February 2026 – Present

Mangaluru, India

Vision and Image Processing Lab, NIT Karnataka

- Developed MedDAMamba, a state-space backbone with dynamic adaptive scanning for CT tumour classification, maintaining accuracy across **2 sites** through domain-adaptive feature learning.

Research Intern, advised by Prof. Pradipta Biswas

May 2025 – July 2025

Bengaluru, India

I3D Lab, Indian Institute of Science (IISc)

- Built a zero-shot synthetic-data pipeline combining diffusion scene editing, CLIP interpolation, and guided inpainting, driving YOLOv8 to **97% and 99.5% mAP** on **2 industrial benchmarks** without manual annotation.
- Ran a VR-headset user study on the deployed system: task completion time down **21.4%**, workload down **41.9%**, and usability up **33.4%** versus the incumbent text-based workflow.

📖 PUBLICATIONS AND PREPRINTS

- PRIMED: Prior-Informed Missing-Modality MRI Synthesis Needs No Sampler. WACV 2027, under review.
- DB-INR: Dual-Band Implicit Neural Representations for Medical Image Segmentation. WACV 2027, under review.
- TuLaBM: Tumor-Biased Latent Bridge Matching for Contrast-Enhanced MRI Synthesis. arXiv:2603.19386.
- SCALA: Semi-supervised Cascade for Left Atrial Scar, Cavity, and Multi-Structure CT Segmentation. MICCAI 2026 CARE LeftAtrium Workshop, oral (top 15%).
- BEAT: Boundary-aware Efficient Anatomy-Transfer for Multimodal Mitral Valve Segmentation. MICCAI 2026 Medical World Model Workshop.
- FieldFiLM: A Unified Conditional Generator for the MRI Field-Strength Continuum. MICCAI 2026 MRIxFields Workshop.
- RADAR: Acquisition-Adversarial Attention Pooling over a Frozen Chest-Radiograph Foundation Model for Tuberculosis Screening. MICCAI 2026 TREAT-MMTB Workshop.
- Improving Mixed Reality Interaction through Generative AI. Intelligent User Interfaces 2027, under review.

🔗 PROJECTS

Ghost-Text Rescue

Vision-Language Models, Python, HuggingFace

- Built a VLM-based OCR pipeline recovering Devanagari text from legacy PDFs with broken glyph mappings, combining page rasterisation, vision-language inference, and quality-aware routing across **2 paths**.
- Quantified the failure it fixes: silent extraction errors in **10.7% of pages** and **14.6% of public government documents** sampled from Common Crawl.

Pramaan

LLM Agents, Python, FastAPI

- Built an agent answering questions on Indian company filings with page-level citations and **6 verification checks**, using hybrid retrieval and a cross-encoder fine-tuned on free XBRL labels (nDCG@10 **0.75 to 0.99**).
- Generated a quarterly-refreshing benchmark of **1,069 questions** from **248 filings**, graded automatically against XBRL; the agent beats naive RAG by **49 points** on the held-out test split (McNemar $p < 0.001$).

🎓 EDUCATION

B.Tech. Computer Science and Engineering

National Institute of Technology Karnataka, Surathkal

August 2023 – May 2027

Mangaluru, India | CGPA 8.97 / 10

CBSE Class 12

Private International English School

August 2022 – March 2023

Abu Dhabi, UAE | 97.6 / 100

🔧 TECHNICAL SKILLS

🔗 Languages

Python, C++, C, Java, MySQL, PostgreSQL

🔗 Agentic AI

AWS Strands SDK, Amazon Bedrock, LangChain, LangGraph, Claude Agent SDK, GitHub Copilot custom agents, HuggingFace, multi-agent orchestration

🔗 ML / Deep Learning

PyTorch, TensorFlow, scikit-learn, NumPy, Pandas, OpenCV, Weights & Biases

🔗 Cloud / Engineering

AWS, Azure DevOps, Docker, Linux, Git, CI/CD, Slurm, FastAPI, Streamlit

🔗 Generative AI

2D/3D Medical Imaging, Diffusion Models, GANs, Vision Transformers, Vision-Language Models (VLMs), RAG

👥 LEADERSHIP

Executive Member and Project Head, Computer Society

IEEE NITK Student Chapter

January 2024 – Present

Mangaluru, India

- Organized the Computer Society project pipeline across 3 academic years, scoping projects, allocating mentors, and reviewing delivery milestones.
- Mentored **2 teams** to public release: QuantumQuill, a transformer-based storyteller, and PathFinder, a career-discovery platform using temporal graph neural networks.
- Contributed MQTT protocol integration to NITK NeST, an open-source network emulation framework, validated against its protocol suite.